

Лабораторная работа №2

Управление пользователями и группами

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1. 1. Цели и задачи работы

1.1 Цель лабораторной работы

Получить практические навыки управления пользователями и группами в Linux.

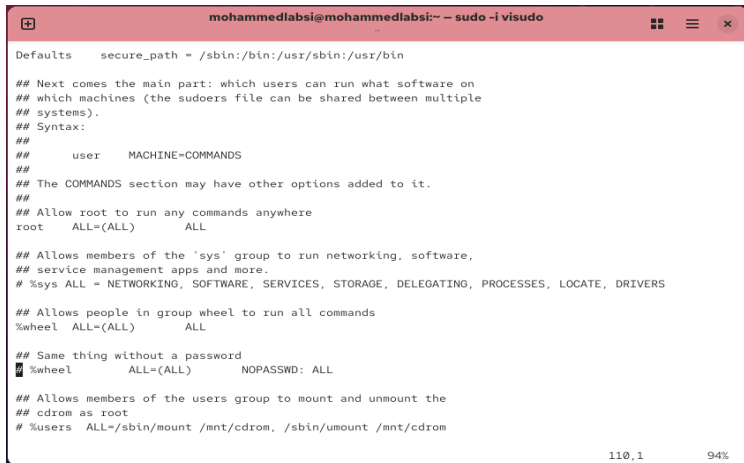
2. 2. Процесс выполнения лабораторной работы

2.1 Определяю текущего пользователя

```
mohammedlabsi@mohammedlabsi:~$ whoami
mohammedlabsi
mohammedlabsi@mohammedlabsi:~$ id
uid=1000(mohammedlabsi) gid=1000(mohammedlabsi) groups=1000(mohammedlabsi),10(wheel) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
mohammedlabsi@mohammedlabsi:~$ su
Password:
root@mohammedlabsi:/home/mohammedlabsi# id
uid=0(root) gid=0(root) groups=0(root) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023
root@mohammedlabsi:/home/mohammedlabsi#
exit
mohammedlabsi@mohammedlabsi:~$
```

Рисунок 1: Определение пользователя и переключение на root

2.2 Просматриваю файл sudoers



```
mohammedlabsi@mohammedlabsi:~ - sudo -i visudo
Defaults    secure_path = /sbin:/bin:/usr/sbin:/usr/bin

## Next comes the main part: which users can run what software on
## which machines (the sudoers file can be shared between multiple
## systems).
## Syntax:
##
##      user    MACHINE=COMMANDS
##
## The COMMANDS section may have other options added to it.
##
## Allow root to run any commands anywhere
root    ALL=(ALL)        ALL

## Allows members of the 'sys' group to run networking, software,
## service management apps and more.
# %sys ALL = NETWORKING, SOFTWARE, SERVICES, STORAGE, DELEGATING, PROCESSES, LOCATE, DRIVERS

## Allows people in group wheel to run all commands
%wheel  ALL=(ALL)        ALL

## Same thing without a password
%wheel  ALL=(ALL)        NOPASSWD: ALL

## Allows members of the users group to mount and unmount the
## cdrom as root
# %users  ALL=/sbin/mount /mnt/cdrom, /sbin/umount /mnt/cdrom
```

110,1 94%

Рисунок 2: Просмотр файла sudoers через visudo

2.3 Создаю пользователей alice и bob

```
mohammedlabsi@mohammedlabsi:~$ sudo -i useradd -G wheel alice
mohammedlabsi@mohammedlabsi:~$ id alice
uid=1001(alice) gid=1001(alice) groups=1001(alice),10(wheel)
mohammedlabsi@mohammedlabsi:~$ sudo -i passwd alice
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
mohammedlabsi@mohammedlabsi:~$ su alice
Password:
alice@mohammedlabsi:/home/mohammedlabsi$ sudo useradd bob

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

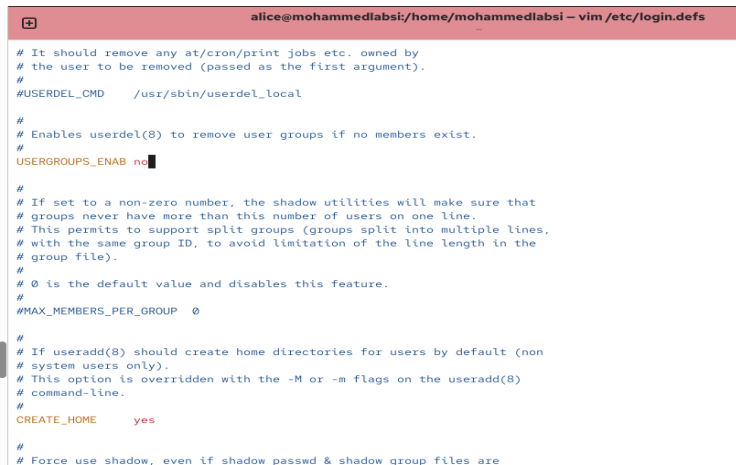
    #1) Respect the privacy of others.
    #2) Think before you type.
    #3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for alice:
alice@mohammedlabsi:/home/mohammedlabsi$ sudo passwd bob
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
alice@mohammedlabsi:/home/mohammedlabsi$ id bob
uid=1002(bob) gid=1002(bob) groups=1002(bob)
alice@mohammedlabsi:/home/mohammedlabsi$ █
```

Рисунок 3: Создание пользователей alice и bob

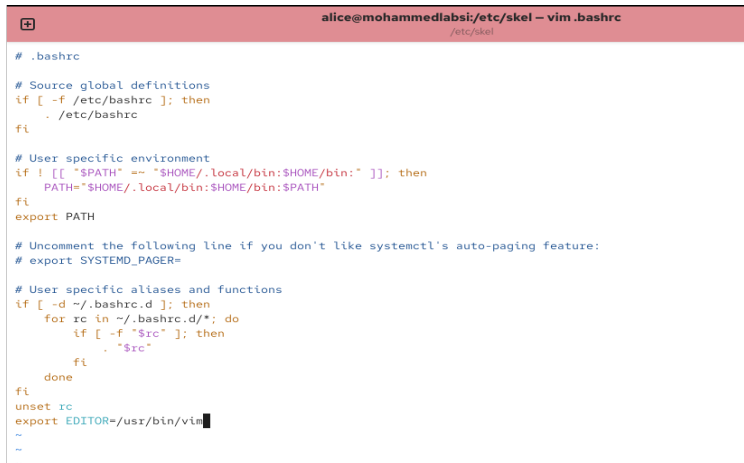
2.4 Настраиваю параметры создания пользователей



```
alice@mohammedlabsi:/home/mohammedlabsi - vim /etc/login.defs
# It should remove any at/cron/print jobs etc. owned by
# the user to be removed (passed as the first argument).
#
#USERDEL_CMD    /usr/sbin/userdel_local
#
# Enables userdel(8) to remove user groups if no members exist.
#
USERGROUPS_ENAB no
#
# If set to a non-zero number, the shadow utilities will make sure that
# groups never have more than this number of users on one line.
# This permits to support split groups (groups split into multiple lines,
# with the same group ID, to avoid limitation of the line length in the
# group file).
#
# 0 is the default value and disables this feature.
#
#MAX_MEMBERS_PER_GROUP 0
#
# If useradd(8) should create home directories for users by default (non
# system users only).
# This option is overridden with the -M or -m flags on the useradd(8)
# command-line.
#
CREATE_HOME      yes
#
# Force use shadow, even if shadow passwd & shadow group files are
```

Рисунок 4: Настройка файла login.defs

2.5 Изменяю шаблонный файл .bashrc



```
alice@mohammedlabsi:/etc/skel - vim .bashrc
/etc/skel

# .bashrc

# Source global definitions
if [ -f /etc/bashrc ]; then
    . /etc/bashrc
fi

# User specific environment
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]; then
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"
fi
export PATH

# Uncomment the following line if you don't like systemctl's auto-paging feature:
# export SYSTEMD_PAGER=

# User specific aliases and functions
if [ -d ~/.bashrc.d ]; then
    for rc in ~/.bashrc.d/*; do
        if [ -f "$rc" ]; then
            . "$rc"
        fi
    done
fi
unset rc
export EDITOR=/usr/bin/vim
~
~
~
```

Рисунок 5: Настройка файла .bashrc в каталоге skel

2.6 Создаю пользователя carol

```
root@mohammedlabsi:/etc/skel#  
root@mohammedlabsi:/etc/skel# su alice  
alice@mohammedlabsi:/etc/skel$ sudo -i useradd carol  
[sudo] password for alice:  
alice@mohammedlabsi:/etc/skel$ sudo passwd carol  
New password:  
BAD PASSWORD: The password is shorter than 8 characters  
Retype new password:  
passwd: password updated successfully  
alice@mohammedlabsi:/etc/skel$ su carol  
Password:  
carol@mohammedlabsi:/etc/skel$ id  
uid=1003(carol) gid=100(users) groups=100(users) context=unconfined_u:unconfined_r:unconfined_t:s0-s0:c0.c1023  
carol@mohammedlabsi:/etc/skel$ cd  
carol@mohammedlabsi:~$ ls -Al  
total 12  
-rw-r--r--. 1 carol users 18 Oct 29 2024 .bash_logout  
-rw-r--r--. 1 carol users 144 Oct 29 2024 .bash_profile  
-rw-r--r--. 1 carol users 549 Sep 12 14:48 .bashrc  
drwxr-xr-x. 2 carol users 6 Sep 12 14:46 Documents  
drwxr-xr-x. 4 carol users 39 Sep 11 19:05 .mozilla  
drwxr-xr-x. 2 carol users 6 Sep 12 14:46 Pictures  
carol@mohammedlabsi:~$
```

Рисунок 6: Проверка пользователя carol и домашнего каталога

2.7 Настраиваю параметры пароля carol

```
carol@mohammedlabsi:~$ su alice
Password:
alice@mohammedlabsi:/home/carol$ sudo cat /etc/shadow | grep carol
carol:$y$j9T$Q2ZLEPp9BzxiDEn7xGzbr/$PoqQEh2rmKkUjrC0cu786eaEp/RfhRZPR/q16ZpiXEC:20708:0:99999:7:::
alice@mohammedlabsi:/home/carol$ sudo passwd -n 30 -w 3 -x 90 carol
passwd: password changed.
alice@mohammedlabsi:/home/carol$ sudo cat /etc/shadow | grep carol
carol:$y$j9T$Q2ZLEPp9BzxiDEn7xGzbr/$PoqQEh2rmKkUjrC0cu786eaEp/RfhRZPR/q16ZpiXEC:20708:30:90:3:::
alice@mohammedlabsi:/home/carol$ sudo grep alice /etc/passwd /etc/shadow /etc/group
/etc/passwd:alice:x:1001:1001:./home/alice:/bin/bash
/etc/shadow:alice:$y$j9T$318Igs15rYhMlj4M4xKN1$e6j7MBompsZouTxBCVwpBjnTifYaaTzn9V/YijvJXo4:20708:0:99999:7:::
/etc/group:wheel:x:10:mohammedlabsi,alice
/etc/group:alice:x:1001:
alice@mohammedlabsi:/home/carol$ sudo grep carol /etc/passwd /etc/shadow /etc/group
/etc/passwd:carol:x:1003:100:./home/carol:/bin/bash
/etc/shadow:carol:$y$j9T$Q2ZLEPp9BzxiDEn7xGzbr/$PoqQEh2rmKkUjrC0cu786eaEp/RfhRZPR/q16ZpiXEC:20708:30:90:3:::
alice@mohammedlabsi:/home/carol$
```

Рисунок 7: Изменение срока действия пароля carol

2.8 Создаю группы и добавляю пользователей

```
alice@mohammedlabsi:/home/carol$  
alice@mohammedlabsi:/home/carol$ sudo groupadd main  
alice@mohammedlabsi:/home/carol$ sudo groupadd third  
alice@mohammedlabsi:/home/carol$ sudo usermod -aG main alice  
alice@mohammedlabsi:/home/carol$ sudo usermod -aG main bob  
alice@mohammedlabsi:/home/carol$ sudo usermod -aG third carol  
alice@mohammedlabsi:/home/carol$ id carol  
uid=1003(carol) gid=100(users) groups=100(users),1004(third)  
alice@mohammedlabsi:/home/carol$ id bob  
uid=1002(bob) gid=1002(bob) groups=1002(bob),1003(main)  
alice@mohammedlabsi:/home/carol$ id alice  
uid=1001(alice) gid=1001(alice) groups=1001(alice),10(wheel),1003(main)  
alice@mohammedlabsi:/home/carol$
```

Рисунок 8: Создание групп main и third

3. 3. Выводы по проделанной работе

В ходе лабораторной работы были изучены команды **whoami**, **id**, **su**, **sudo**, **useradd**, **passwd**, **groupadd** и **usermod**.

Были созданы пользователи **alice**, **bob** и **carol**, настроены права администратора через группу **wheel**, изменены параметры создания новых пользователей и выполнена работа с группами **main** и **third**.